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Influence of Background Speech and Music in Interrupted Reading: An Eye-Tracking Study

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Summary: The current study examined the influence of interruption, background speech and music on reading, using an eye movement paradigm. Participants either read paragraphs while being exposed to background speech or music or read the texts in silence. On half of the trials, participants were interrupted by a 60-second audio story before resuming reading the paragraph. Interruptions increased overall reading time, but the reading of text following the interruption was quicker compared with baseline. Background speech and music did not modulate the interruption effects, but the background speech slowed down the reading rate compared with reading in the presence of music or reading in silence. The increase in reading time was primarily due to an increase in the time spent rereading previously read words. We argue that the observed interruption effects are in line with a theory of long-term working memory, and we present practical implications for the reported background speech effects.
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INTRODUCTION

Reading a text is an everyday task that takes place in various contexts and is always prone to a host of potential distractions. Office workers are exposed to the conversations of their neighbours and are frequently interrupted to answer a 'quick' question. Many people also deliberately listen to music while they read—often in an effort to immunize themselves against more substantial disruptions. Given that reading is a complex task that requires the maintenance of a large amount of information in memory, one may wonder how people can cope with such distractions. Although the influence of interruptions (e.g. Glanzer, Fischer, & Dorfman, 1984), background speech (e.g. Banbury & Berry, 1998) and music (e.g. Furnham & Strbac, 2002) on reading has been extensively studied, none of these studies have used the eye-tracking technique (Rayner, 1998) to gain insight into the addressed research issues.

The goal of the present study is to make use of the advantages of eye-tracking to re-examine the influence of interruptions, background speech and music on the reading process. We will briefly review previous research on interruption effects during reading and illustrate how this area of research can benefit from the use of eye-tracking techniques. Then we examine how eye-tracking techniques can help us to investigate the effect of background speech and music on reading.

In a series of experiments, Glanzer and collaborators (Fischer & Glanzer, 1986; Glanzer, Dorfman, & Kaplan, 1981; Glanzer et al., 1984) presented their participants with a text of a few sentences in either a non-interrupted or an interrupted format. Using a self-paced reading method, participants read one sentence at a time. The interruptions introduced arithmetic or unrelated reading tasks in between two sentences of the primary text. Glanzer and collaborators found that although overall comprehension was not affected, the reading time of the first sentence following the interruption was inflated relative to the uninterrupted condition. This effect disappeared when the readers were presented with the last one or two sentences that preceded the interruption before reading the post-interruption sentences. A subsequent study (Lorch, 1993) showed that the effect could also disappear when the reader was reminded of the general topic of the text before reading the post-interruption sentences. These findings show that in order to integrate the content of successive sentences into a coherent representation, readers keep information about previously read text accessible in working memory. The maintained information includes the local context as well as more integrated information such as the general topic of the text.

Ericsson and Kintsch (1995) interpreted these findings in the context of their Long-Term Working Memory theoretical framework. According to this framework, in normal reading, end products of processing can be rapidly stored in and retrieved from long-term memory so that long-term memory serves as an extended working memory. Information about previous text stored in long-term memory is kept accessible by means of retrieval cues stored in short-term working memory. The findings reported by Lorch and Glanzer and collaborators support the Long-Term Working Memory account mainly for the following reason: although interruptions disrupt the information stored in short-term working memory during reading, it does not appear to cause an irretrievable loss of information, as the overall comprehension of the text is not impaired. Thus, information seems to be rapidly and accurately consolidated into long-term memory during reading. According to Ericsson and Kintsch (1995), when readers resume reading the text after the interruption, the novel information serves as a retrieval cue to information about text previously read and stored in long-term memory, and observed increased reading times reflect these retrieval operations. This interpretation was supported more recently by a study showing that interruptions do not hamper comprehension regardless of their intensity and difficulty, unless the interface drastically limits the reading time so that the readers do not have enough time to store information in long-term memory (Oulasvirta & Saariluoma, 2006).

Without doubt, studies by Lorch and Glanzer and collaborators have substantially advanced our understanding of the role played by memory processes in reading. However,

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because they employed the sentence-by-sentence presentation procedure, they naturally reveal little about how interruptions affect normal reading. Participants in these studies were always artificially constrained to resume reading at a particular location in the text. Because of this, the effects of interruptions during normal reading are still unclear. For instance, when individuals resume reading a text, do they normally reread the text located before the interruption point? And if so, to what extent? How do readers process the pre-interruption or the post-interruption text? Do they spend smaller, equal or greater amount of time reading the post-interruption text compared with a no-interruption situation? Allowing several lines of text to be displayed at the same time, eye-tracking is the only method to be in a position to provide answers to such questions. Because it does not disrupt normal reading, it permits a realistic investigation of the influence of interruptions on normal reading.

The influence of background speech and music on cognitive performance is a long-standing issue that still motivates substantial research. A number of studies have consistently shown that background speech severely impairs immediate memory for visually presented verbal material (Beaman & Jones, 1998; Salamé & Baddeley, 1982, 1989)—an effect known as the *irrelevant* or *unattended speech* effect. Subsequent research extended this effect in the context of a reading comprehension task, showing that overall comprehension was clearly impaired in the presence of background speech relative to reading in silence (Banbury & Berry, 1998; Martin, Wolgalter, & Forlano, 1988; Sorqvist, Halin, & Hygge, 2010), with meaningful speech being more disruptive than meaningless speech (Oswald, Tremblay, & Jones, 2000). According to the interference-by-process account of auditory distraction (Marsh, Hughes, & Jones, 2008, 2009), background speech disrupts performance on tasks such as reading comprehension because of interference between two sets of processes: the semantic processes engaged in the reading task on the one hand and automatic processing of semantic information on the other. Yet, because these studies merely used overall comprehension measures, they tell us little about the way the semantic processes involved in the reading task are disrupted by background speech. For instance, the influence of concurrent speech on overall reading rate, if any, is still unknown.

Eye-tracking seems an appropriate methodology to examine the way in which the reading process is disrupted by concurrent speech because it provides an excellent indication of the moment-to-moment cognitive processes underlying reading. Reading requires the identification of words and the integration of this lexical information into the overall meaning of the sentence and the text as a whole (i.e. post-lexical processing). Eye movements have proved to reveal processing difficulties at both lexical and post-lexical processing levels. For instance, low-frequency words are known to yield longer fixation durations (Inhoff & Rayner, 1986), whereas words of which meaning do not fit well within their semantic context (e.g. implausible words) yield longer fixation durations and backward eye movements (i.e. regressions) (Warren & McConnell, 2007). Computational models of eye movement control during reading have traditionally focused only on low-level processes related to lexical processing (see Engbert, Nuthmann, Richter, & Kliegl, 2005, for a review). However, a recent version of the E-Z Reader model (E-Z Reader 10; Reichle, Warren, & McConnell, 2009) takes into account both lexical and post-lexical processes. The model builds on a large body of research results, and one of the important assumptions that are made is that lexical processing difficulties should primarily cause longer fixation durations, whereas higher-level post-lexical processing difficulties should cause both longer fixation durations and backward eye movements.

Although previous research clearly indicates that background speech impairs reading comprehension, no such unequivocal conclusion has been reached for background music. Some studies reported a decrease in comprehension scores in the presence of concurrent music (Fogelson, 1973; Furnham & Strbac, 2002; Kiger, 1989), whereas others failed to find any reliable effects (Furnham & Allass, 1999; Furnham & Bradley, 1997; Furnham, Trew, & Sneade, 1999; Martin et al., 1988; Tucker & Bushman, 1991). Research has also shown that effects of music on reading comprehension depend on type of music (e.g. Kiger, 1989) and personality traits (e.g. Furnham & Bradley, 1997) and are likely to be in part a consequence of differences in arousal and mood rather than a consequence of the music per se (e.g. Thompson, Schellenberg, & Husain, 2001). For instance, on the basis of the large volume of research showing a clear impairment in reading comprehension in the presence of background speech, one can expect vocal music to be generally more disruptive than instrumental music. Accordingly, Salamé and Baddeley (1989) found that vocal music significantly impaired performance on an immediate recall task, whereas instrumental music only disrupted recall to a far lesser extent. As with the research relating to background speech, despite the large number of studies carried out, no study has ever made use of the eye-tracking technique to examine the effect of concurrent music on reading comprehension.

The goal of the present study was to make use of the eye-tracking technique to investigate (i) how interruptions exert their effects on the reading process in a normal reading situation; (ii) how auditory distractions caused by background speech and music affect the reading process; and (iii) whether background speech and music amplify the deleterious effects of interruptions. Participants read short paragraphs while their eye movements were recorded. On half of the trials, they were interrupted by a 60-second audio story before resuming reading the paragraph. They read the paragraphs in the presence of background speech, instrumental music (played at the same sound level) or in silence. After having read each paragraph, participants answered a comprehension question and provided subjective rating of difficulty of comprehension.

For the interruption effects, we made the following predictions. We expected that when resuming reading the paragraphs, the readers would substantially reread the pre-interruption text to reinstate the corresponding information in working memory and find the entry point of the novel text portion. We predicted an increase in overall reading time because of this rereading strategy. On the basis of previous studies of repeated reading (Hyöna & Niemi, 1990; Raney & Rayner, 1995; Raney, Therriault, & Minkoff, 2000) that

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found readers to make fewer fixations, longer saccades and have shorter gaze durations when rereading a text, we predicted fewer fixations, longer saccades and shorter gaze durations on the pre-interruption text after the interruption than before the interruption, indicating a skimming fixation pattern. On the basis of previous interruption studies (Fischer & Glanzer, 1986; Glanzer, et al., 1981, 1984; Lorch, 1993) that showed that interruption effects disappeared if readers were allowed to reread the pre-interruption text regions (e.g. the last sentence) when resuming reading, we predicted no inflated reading times on the post-interruption text compared with the no-interruption baseline condition. Finally, in accordance with the previous empirical studies into interruption effects and the Long-Term Working Memory theoretical framework (Ericsson & Kintsch, 1995), we expected no decrease in comprehension performance in the interruption condition compared with the no-interruption condition.

With regard to the effect of background speech and music, we hypothesized the following: an extensive body of research showed that reading comprehension is greatly impaired in the presence of background speech (e.g. Banbury & Berry, 1998). In contrast, music and especially instrumental music seems to have a much more limited effect on reading comprehension, if any (Martin et al., 1988). Therefore, we expected background speech, but not instrumental music, to impair the comprehension performance of the participants, compared with the silence baseline condition. Importantly, we also expected background speech, but not instrumental music, to affect the various aspects of eye movement behaviour, as eye movements are a sensitive indicator of cognitive processing in reading. More specifically, we predicted longer gaze durations, shorter saccades, more fixations, higher regression probabilities and longer reading times in the background speech condition compared with the silence and instrumental music conditions, reflecting a general disruption of the reading process at both lexical and post-lexical stages.

Finally, if background speech severely impairs reading comprehension, it should prevent the reader from accurately storing information about text previously read in long-term memory, or it should at least prevent the reader from storing that information in a well-organized and structured form. As a result, information about text previously read should not be available or easily accessible when resuming reading a text after an interruption, further increasing the difficulty of the reading resumption process. In sum, we expected the deleterious effects of interruption being magnified in the presence of background speech, but not instrumental music, leading to an additional increase in overall reading time.

METHOD

Participants

Thirty-two University of Kent students participated in the study for course credit or payment. They were all native English speakers with normal uncorrected vision. Two participants had to be excluded because of poor calibration; as a consequence, all subsequent analyses are based on a sample of 30 participants. There were 21 women and 9 men with an age range of 18–29 years ($M = 21.7$, $SD = 2.6$). All participants were treated in line with the ethical codes laid down by the University of Kent and the ethical considerations put forward by the British Psychological Society.

Apparatus

Eye movements were recorded using an EyeLink 1000 desktop mounted SR Research eyetracker (see http://sr-research.com/pdf/techspec.pdf). Sampling frequency was 1000 Hz; only the right eye was tracked. The participant's head was kept immobile with the use of a chin and head rest. The stimuli were presented on a 19-inch CRT screen with a screen resolution of 1024×768 pixels, at a distance of 60 cm from the participant.

MATERIALS

The reading material consisted of 24 paragraphs that were adapted from a number of different textbooks, biographical texts and fiction texts. They were selected such that they did not require any domain-specific knowledge and were relatively easy to understand on first reading (see the Appendix for an example paragraph). The paragraphs were presented double-spaced on the computer screen and occupied one screen containing 14 lines of text. They were on average 158.35 ($SD = 6.80$) words in length. For each paragraph, a target word within line 7 was chosen. Upon moving the eyes over this word, the interruption message was triggered. Target words were chosen at random, but words at the beginning and end of the line were avoided. The randomization led to the target words being the first word of a sentence two times over 24, resulting in almost all of the interruptions occurring while the participants were processing a sentence rather than at the juncture between two sentences. In almost all of the previous interruption studies, the interruptions always occurred at the juncture between two sentences. However, a study conducted by McNamara and Kintsch (1996) showed that such interruptions are less disruptive than interruptions occurring in the middle of a sentence. Moreover, one can argue that there is a low probability of interruptions occurring exactly at the juncture between two sentences in natural settings. Therefore, our selected target words deliberately led to interruptions that were more representative of interruptions in natural settings compared with previous studies.

One comprehension question was created for each paragraph. The questions took the form of multiple choice questions, and each question was presented with two possible answers, which could both be considered perfectly feasible if the paragraph had not been read previously. Van Dijk and Kintsch (1983) have distinguished among three levels of text representation: the surface structure (representation of the exact wording of the text), the textbase (representation of the semantic content of the text) and the situation model (integration of the text content with the reader's world knowledge). As for the previous interruption (e.g. Glanzer et al., 1984; Oulasvirta & Saariluoma, 2006) and background speech studies (e.g. Banbury & Berry, 1998; Oswald et al., 2000), answering the questions required recall of local fragments of the text and did not require inference (see the Appendix for an example question). Accordingly, the questions tapped into the surface and the textbase levels of representation. The text fragments

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that needed to be recalled were located before the interruption point (11 questions), after the interruption point (6 questions) or both before and after the interruption point (7 questions).

Interruptions took the form of audio news stories that were 60 seconds in length. These news stories were taken from UK news websites (e.g. BBC news, Guardian Online, Times Online). The stories were read into a dictaphone in the same male voice and were subsequently converted into digital. wav files for the experiment. In total, 15 audio files were used.

In the background speech and background music conditions, a single audio file was used for each. The background speech file was taken from a radio programme called 'between the lines' on National Public Radio, which consisted of conversations between book writers and the radio host. The file lasted 29 minutes. The music file consisted of the first five songs of an album called The Exquisite Death of Saxon Shore, from the rock band Saxon Shore. The music file lasted 24 minutes in total and was entirely instrumental. The main criterion for selecting the speech and music files was that they would not be familiar to the participants (which the experimenter ensured while running the experiment). In doing so, we circumvented the well-known effect that participants' episodic memory is activated when listening to familiar audio pieces (Eschrich, Munte, & Altenmuller, 2008), which could have introduced a bias in the music manipulation. Both background speech and music were played at the same sound level (i.e. at approximately 60–70 decibels).

Procedure

The experiment began with a calibration of the eyetracker by means of a nine-point calibration grid that covered the entire computer screen. After a familiarization phase, the participants read 24 paragraphs. The participants were instructed to read the paragraph for comprehension as they would be asked questions about them afterwards. After each paragraph, they were immediately presented with a comprehension question that they responded to by using the keyboard. They were then asked to rate the difficulty of the paragraph on a seven-point scale (1 being very easy and 7 being very difficult).

On half of the paragraphs, when the gaze of the reader landed on a pre-defined target word, the text disappeared and a 60-second audio story was played via headphones that the participant was wearing. Following the audio story, a blue box appeared on the computer screen, and the participant was to describe briefly the story in a sentence to the experimenter. The experimenter ensured that the description phase would never exceed a very brief period of time (approximately 5–7 seconds in maximum). The participant was then presented with a cross to fixate at the top of the screen, and then returned to the paragraph he or she was reading previously so that he or she could continue reading. For the other half of the paragraphs (i.e. in the no-interruption condition), no audio story interrupted the reading of the paragraph when the gaze of the reader landed on the pre-defined target word. However, a virtual mark was added to the data in order to facilitate data analysis (so that pre-interruption and post-interruption phases were distinguished in interruption and no-interruption conditions, see Results section). Once the participants had finished reading the paragraph, they pressed the 'space bar' and the paragraph was removed from the screen. During the familiarization phase of the experiment, the participants were made aware that the screen may go blank during reading and that they may hear an audio story through headphones and be asked to briefly describe the audio story in a sentence afterwards. They were informed that following the story, they were to finish reading the text. They were also advised that they were free to resume reading the texts as they wished, as long as they were able to answer the comprehension question afterwards.

In addition to the interruption manipulation, the participants read successively one third of the paragraphs in the presence of background speech, one third in presence of music and one third in silence (the order of presentation was counterbalanced across participants, see below). The background audio files were played via remote speakers connected to an independent computer, and the participants were informed of this manipulation during the familiarization phase. They were instructed to focus on the reading material and the audio stories, while not paying attention to the auditory stimuli coming from the speakers.

The paragraphs were presented in a fixed order. The distribution of the paragraphs to the interruption and no-interruption conditions was counterbalanced across participants, and the order of presentation of the audio stories was randomized for each participant. The order of presentation of the background speech, music and silence conditions was counterbalanced across participants in a 3×3 Latin square design.

RESULTS

To analyse the effects of interruptions, background speech and music, we computed separate 2 (interruption: interruption, no-interruption) $\times$ 3 (background sound: speech, music, silence) repeated measures analyses of variance (ANOVAs). Interruption and background sound were within-participant factors. Dependent measures included several eye movement measures, a measure of reading comprehension and a subjective measure of ease/difficulty of comprehension. The eye movement measures were the reading time (i.e. the summed duration of fixations), the gaze duration (i.e. the summed duration of fixations prior to exiting a word for the first time), the number of fixations, the probability of regression and the length of saccades.

Initially, we present the results of the interruption effects. Next, we report the results regarding the effects of background speech and music and the potential interactions with the interruption manipulation.

Effects of interruptions

Three main sets of analyses were conducted on the eye movement measures, in accordance with the selected regions and periods of interests. The data are illustrated in Table 1. We distinguished the pre-interruption and post-interruption periods, depending on whether the eyes of the reader had or had not already landed on the pre-defined target word (the word triggering an interruption in the interruption trials). In addition, we distinguished the pre-interruption and post-interruption

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Table 1. Eye movement results for the pre-interruption and post-interruption text (spacewise) and for the pre-interruption and post-interruption periods (timewise) as a function of the background sound conditions

		Pre-interruption period		Post-interruption period			
		Pre-interruption text		Pre-interruption text		Post-interruption text	
		No interruption	Interruption	No interruption	Interruption	No interruption	Interruption
Reading time (seconds)	Silence	15.81 (0.75)	16.33 (0.80)	0.60 (0.11)	5.75 (0.74)	18.22 (0.99)	17.05 (1.13)
	Music	15.93 (0.67)	16.14 (0.81)	0.75 (0.19)	4.65 (0.53)	18.25 (1.09)	17.82 (0.95)
	Speech	18.47 (0.98)	17.88 (0.93)	0.51 (0.09)	5.42 (0.59)	20.15 (1.21)	18.66 (1.15)
Number of fixations	Silence	73.24 (2.65)	75.38 (2.95)	2.88 (0.60)	30.20 (3.84)	82.57 (3.77)	80.99 (4.45)
	Music	74.20 (2.40)	74.73 (3.30)	3.78 (1.05)	24.48 (2.84)	84.05 (4.35)	84.65 (3.89)
	Speech	82.10 (3.24)	81.16 (3.42)	2.50 (0.41)	26.96 (2.89)	88.98 (4.13)	86.75 (4.26)
Gaze duration (milliseconds)	Silence	249.42 (7.38)	251.11 (6.93)	269.16 (21.44)	211.34 (5.46)	254.60 (7.72)	239.16 (7.22)
	Music	244.35 (6.94)	249.34 (7.60)	218.16 (9.93)	207.40 (6.79)	248.07 (7.64)	240.26 (7.04)
	Speech	257.41 (8.44)	256.07 (7.99)	243.31 (15.36)	225.87 (9.32)	265.40 (8.44)	246.14 (8.13)
Probability of regressions (%)	Silence	0.18 (0.01)	0.18 (0.01)	0.11 (0.02)	0.17 (0.01)	0.18 (0.01)	0.19 (0.01)
	Music	0.20 (0.01)	0.20 (v.01)	0.11 (0.02)	0.19 (0.01)	0.19 (0.01)	0.20 (0.01)
	Speech	0.20 (0.01)	0.19 (0.01)	0.15 (0.02)	0.16 (0.01)	0.19 (0.01)	0.20 (0.01)
Saccade length (°)	Silence	3.67 (0.10)	3.70 (0.10)	5.54 (0.83)	4.28 (0.15)	3.69 (0.13)	3.87 (0.14)
	Music	3.75 (0.12)	3.80 (0.12)	6.06 (0.77)	4.38 (0.14)	3.77 (0.11)	3.88 (0.12)
	Speech	3.69 (0.10)	3.71 (0.14)	5.26 (0.80)	4.23 (0.14)	3.64 (0.12)	3.82 (0.14)

All values are presented as means (and standard errors).

text (i.e. the text located before the target word and the text located after the target word). Accordingly, the first set of analyses concerned the initial reading of the pre-interruption text, before the reader had reached the target word at least once. The second set of analyses related to the pre-interruption text after the reader already reached the target word (and triggered an interruption on the interruption trials). The third set of analyses concerned the post-interruption text (and consequently the post-interruption period).

Each trial was checked to see if the eyes of the reader had reached the target word at an appropriate time. Only trials in which the target word had been reached after the previous lines had been covered, and in which the target word was reached prior to the post-interruption text being read, were included in the final analyses. These stringent criteria led to 12% of erroneous data being removed but were necessary to avoid trials with a premature delivery of the interruption text to be included. Analyses were conducted on the remaining data.

The first set of analyses performed on the eye movement data collected during the pre-interruption period (on the pre-interruption text) did not yield any reliable effect of the interruption factor (see Table 1). This was not surprising as interruption and no interruption trials were completely unpredictable. Irrespective of the interruption manipulation, the participants spent on average 16.76 seconds reading the pre-interruption text during this period, making 76.80 fixations, with a mean gaze duration of 251.28 milliseconds, a mean regression probability of.19 and a mean saccade length of 12.54 character spaces.

As expected, the second set of analyses performed on the eye movement data showed that the participants spent substantial time rereading the pre-interruption text when returning to the paragraphs after an interruption. They spent on average 5.27 seconds rereading the pre-interruption text, which contrasted clearly with the mean 0.62 seconds spent rereading the pre-interruption text after having reached the target word in the no interruption trials, $F(1, 29) = 101.33$, $\mathrm{MSE} = 9.60$, $p < .01$, $\eta_p^2 = .77$ (in the latter condition, rereading was due to incidental regressions often occurring during reading; MSE stands for Mean Square Error). Accordingly, the readers made 27.21 fixations on average in the interruption trials and 3.05 fixations in the no interruption trials, $F(1, 29) = 107.14$, $\mathrm{MSE} = 245.16$, $p < .01$, $\eta_p^2 = .79$. As predicted, comparison of the characteristics of the eye movements collected on the pre-interruption text after the interruption with the characteristics of the eye movements collected before the interruption on the same text portion revealed a skimming fixation pattern; reading time was much shorter (5.27 versus 16.78 seconds), there were much fewer fixations (27.21 versus 77.09), gaze durations were much shorter (214.87 versus 252.17 milliseconds) and saccades were longer (14.49 versus 12.60 characters). A separate analysis of the eye movement data for each line of the pre-interruption text further characterized this skimming fixation pattern. The analysis revealed that most of the fixations the readers made were located near the last line of the pre-interruption text: Within lines 1 through 7, readers made on average 2.9, 2.67, 2.48, 3.22, 4.48, 7.15 and 10.47 fixations, respectively. Moreover, the first lines of text were skipped by a large proportion of the readers: for lines 1 through 7, the lines were skipped on average by 37%, 38%, 55%, 45%, 30%, 9% and 1% of the readers, respectively.

Opposite to our expectations, the third set of analyses yielded a reliable effect of interruption on three of the five eye movement measures. We expected no difference in reading time for the post-interruption text between the interruption and no interruption trials. Yet, the analyses revealed that participants spent *less* time reading the post-interruption text in the interruption trials ($M = 17.84$ seconds) compared with the no interruption trials ($M = 18.87$ seconds), $F(1, 29) = 11.59$, $\mathrm{MSE} = 4.13$, $p < .01$, $\eta_p^2 = .29$. A subsidiary analysis showed that this increase in reading time was constant over the entire post-interruption text. This effect was associated with shorter gaze durations (241.85 versus

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256.02 milliseconds), $F(1, 29) = 41.19$, MSE = 219.24, $p < .01$, $\eta_p^2 = .59$; and longer saccade lengths (13.01 versus 12.47 characters), $F(1, 29) = 19.90$, MSE = 0.05, $p < .01$, $\eta_p^2 = .40$. These unexpected changes in eye movement behaviour suggest that the processing of the post-interruption text was somehow *facilitated* in the interruption trials compared with the no interruption trials. An additional analysis of the total time spent reading the text, irrespective of the selected regions/periods of interest, yielded the following result. Although the processing of the post-interruption text was accelerated in the interruption condition, the substantial amount of time spent rereading the pre-interruption text when resuming reading after an interruption still led to an increase in overall reading time in the interruption condition (39.91 versus 36.25 seconds), $F(1, 29) = 32.11$, MSE = 18.78, $p < .01$, $\eta_p^2 = .53$.

Finally, the data corresponding to the comprehension and subjective rating measures are illustrated in Table 2. Consistent with our expectations, the analysis performed on the reading comprehension measure suggested that the overall comprehension of the paragraphs was not affected by the interruptions. The comprehension scores were even (but not significantly) higher in the interruption condition ($M = 0.88$) than in the no interruption condition ($M = 0.86$). Although these results should be considered with caution given the possibility of a ceiling effect (note though that Glanzer and collaborators obtained similar scores), they suggest that the interruptions did not drastically affect the overall comprehension of the paragraphs. The analysis performed on the subjective difficulty ratings showed that participants reported paragraphs as being more difficult to understand in the interruption condition ($M = 3.28$) compared with the no interruption condition ($M = 3.02$), $F(1, 29) = 5.84$, MSE = 0.51, $p < .05$, $\eta_p^2 = .17$. This last result is not surprising considering that the participants spent on average 10% extra time processing the paragraphs in the interruption condition compared with the no interruption condition (see above).

Effects of background speech and music

Next, we consider the potential interaction between the interruption and background sound factors. Finally, we will present the results related to the main effects of background speech and music.

Our expectations were that background speech, but not music, would amplify the deleterious effects of interruptions. Contrary to these expectations, the analysis performed on the total reading time showed that the increases in reading time caused by the interruptions were no different in the presence of background speech, music or silence, as indicated by a lack of an interruption × background sound interaction (see Table 1). The costs associated with the interruptions were even (but not significantly) *smaller* in the speech and music conditions ($M = 2.82$ seconds and $M = 3.67$ seconds, respectively) compared with the silence condition ($M = 4.49$ seconds). Furthermore, the analyses performed on all of the dependent variables did not yield any interruption × background sound interaction. This suggests that neither background speech nor music amplify the deleterious effects of interruptions. However, as we will see in the following sections, these results by no means indicate that background speech cannot have deleterious effects on the reading process, but rather that the effects of interruptions on the one hand and the effects of background speech on the other hand are simply additive and independent.

Table 2. Comprehension and difficulty rating scores as a function of the interruption and background sound conditions

		No interruption	Interruption
Comprehension scores	Silence	0.87 (0.03)	0.90 (0.02)
	Music	0.91 (0.02)	0.86 (0.03)
	Speech	0.82 (0.04)	0.89 (0.04)
Difficulty ratings	Silence	2.55 (0.19)	3.13 (0.16)
	Music	2.88 (0.22)	3.00 (0.19)
	Speech	3.63 (0.22)	3.71 (0.22)

All values are presented as means (and standard errors).

Given that the background sound factor did not interact with the interruption factor, the data of the pre-interruption and post-interruption regions/periods of interest were aggregated and one-way ANOVAs with three levels (background speech, music and silence) were performed on all of the eye movement measures for the no interruption trials only. The analyses showed a reliable effect of background sound for three of the five eye movement measures, namely the reading time, $F(2, 28) = 12.26$, MSE = 15.48, $p < .01$, $\eta_p^2 = .30$; the number of fixations, $F(2, 28) = 6.20$, MSE = 294.66, $p < .01$, $\eta_p^2 = .18$; and the gaze duration, $F(2, 28) = 11.18$, MSE = 160.27, $p < .01$, $\eta_p^2 = .28$. Post-hoc tests showed that, consistent with our expectations, the background speech condition differed in each case with the silence and music conditions, whereas the silence and music conditions never differed significantly. Consistent with our predictions, the participants spent more time reading the paragraphs (39.14 versus 34.65 and 34.95 seconds), made more fixations (173.58 versus 158.70 and 162.03) and had longer gaze duration (261.61 versus 252.21 and 246.28 milliseconds) in presence of background speech compared with the silence and music conditions, respectively (all ps $< .05$, post-hoc tests with Bonferroni correction), indicating a clear disruption of the reading process in presence of background speech. However, as opposed to our predictions, the probability of regression and the saccade length measures did not yield any reliable differences.

As interword regressions are generally assumed to reflect high-level processing difficulty during reading, it was surprising to note a lack of clear increase in the probability of regression in the presence of background speech. We reasoned that one potential cause for this null result could have been that the chosen measure was not sensitive enough to detect any effect. Therefore, we computed additional measures related to regressions, that is, one word-level measure and three sentence-level measures. The additional word-level measure was the rereading time, which corresponded to the time spent fixating words already accessed and exited at least once. The sentence-level measures were the first-pass forward fixation time, the first-pass rereading fixation time and the look-back fixation time (see Hyönä, Lorch, & Kaakinen, 2002; Hyönä, Lorch, & Rinck, 2003). The first-pass forward fixation time corresponded to the time spent fixating previously unread portions of the sentences during initial reading (i.e. before exiting the sentences at least once). The first-pass rereading fixation time corresponded to the time spent fixating previously read

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portions of the sentences during initial reading. The look-back fixation time corresponded to the time spent fixating sentences already accessed and exited at least once. In sum, the rereading time measure was indicative of interword regressions, the first-pass rereading fixation time was indicative of intrasentence regressions and the look-back fixation time was indicative of intersentence regressions. The data are presented in Table 3.

The analysis performed on the rereading time measure showed that the participants spent markedly more time rereading previously read words in the presence of background speech ($M = 10.03$ seconds, $SE = 0.97$) compared with the silence condition ($M = 7.42$ seconds, $SE = 0.70$) and the music condition ($M = 8.19$ seconds, $SE = 0.75$), {all $ps < .05$, post-hoc tests performed with Bonferroni correction after a significant one-way three-level ANOVA [$F(2, 28) = 8.80$, MSE = 6.14, $p < .01$, $\eta_p^2 = .23$]}. It should be noted that the increase in rereading time in the background speech condition compared with the silence condition (i.e. 2.61 seconds) accounted for 58% of the increase in overall reading time (i.e. 4.49 seconds). The analysis performed on the first-pass forward fixation time measure yielded a reliable effect of background sound, $F(2, 28) = 5.67$, MSE = 6.72, $p < .01$, $\eta_p^2 = .16$, but post-hoc tests indicated that the only comparison that proved significant was the comparison between the music condition ($M = 27.04$ seconds, $SE = 1.39$) and the background speech condition ($M = 29.30$ seconds, $SE = 1.64$). The results obtained with the first-pass rereading fixation time measure mimicked the results obtained with the rereading time measure: the ANOVA proved significant [$F(2, 28) = 10.86$, MSE = 4.95, $p < .01$, $\eta_p^2 = .27$], and post-hoc tests showed that participants spent more time rereading portions of the sentences during initial reading in the background speech condition ($M = 9.20$ seconds, $SE = 0.93$) than in the silence condition ($M = 7.02$ seconds, $SE = 0.60$); $p < .05$, post-hoc tests performed with Bonferroni correction. This increase in first-pass rereading time (i.e. 2.18 seconds) accounted for 48% of the increase in overall reading time. The analysis performed on the look-back fixation time measure yielded also a reliable effect of the background sound factor, $F(2, 28) = 4.12$, MSE = 3.46, $p < .05$, $\eta_p^2 = .12$, and post-hoc tests indicated that the participants spent more time returning to previous sentences in the background speech condition ($M = 2.41$ seconds, $SE = 0.42$) than in the silence condition ($M = 1.18$ seconds, $SE = 0.20$); $p < .05$, post-hoc tests performed with Bonferroni correction. This increase in look-back fixation time (i.e. 1.23 seconds) accounted for 27% of the increase in overall reading time.

Table 3. Additional analyses of the eye movement data as a function of the background sound conditions

	Word-level measures	Sentence-level measures		
	Rereading time	First-pass forward fixation time	First-pass rereading fixation time	Second-pass fixation time
Silence	7.42 (0.70)	28.24 (1.53)	7.02 (0.60)	1.18 (0.20)
Music	8.19 (0.75)	27.04 (1.39)	6.76 (0.62)	2.33 (0.41)
Speech	10.03 (0.97)	29.30 (1.64)	9.20 (0.93)	2.41 (0.42)

All values are presented as means (and standard errors) in seconds.

Surprisingly, the post-hoc tests also revealed that the participants spent more time returning to previous sentences in the music condition ($M = 2.33$ seconds, $SE = 0.41$) than in the silence condition, $p < .05$. But it should be recalled that this increase in look-back fixation time in the music condition compared with the silence condition did not lead to an increase in overall reading time (see above).

The analysis performed on the comprehension measure did not yield any reliable effects of the background sound factor. The data are presented in Table 2. In particular, although the difference in the means went in the predicted direction, comprehension scores observed in the background speech condition ($M = 0.82$) did not differ significantly from the scores observed in the silence condition ($M = 0.87$), but as stated previously, the results of the comprehension test should be considered with caution given the possibility of a ceiling effect. On the contrary, the analysis performed on the subjective difficulty ratings yielded a main effect of background sound, $F(2, 28) = 8.62$, MSE = 1.05, $p < .01$, $\eta_p^2 = .23$. Post-hoc tests showed that the participants reported the paragraphs being more difficult to understand in the background speech condition ($M = 3.63$) compared with the silence condition ($M = 2.55$) and the music condition ($M = 2.88$), (all $ps < .05$, post-hoc tests with Bonferroni correction). This last result was not surprising given that the eye movement measures indicated a clear disruption of the reading process.

DISCUSSION

The first goal of the present study was to investigate how readers cope with interruptions in a normal reading situation. The major findings can be summarized as follows. The interruptions led to an increase in overall reading time because of readers rereading the pre-interruption text when resuming reading after the interruption. The processing of the post-interruption text proved to be *facilitated* by the interruption, as indicated by a decrease in reading time in particular. Finally, although results of the comprehension test should be considered with caution given the possibility of a ceiling effect, the interruptions did not seem to affect overall comprehension. In the following, we will discuss the implication of these results in the context of previous interruption studies and argue that they are most in line with the Long-Term Working Memory theory. We will then present and discuss the major findings related to the second and third goals of our study, namely to investigate the effects of background speech and music on the reading process and to examine whether these factors can modulate the effects of interruptions.

As pointed out in the introduction section, previous interruption studies (Fischer & Glanzer, 1986; Glanzer, et al., 1981, 1984; Lorch, 1993; McNamara & Kintsch, 1996; Oulasvirta & Saariluoma, 2006) have greatly increased our understanding of the effects of interruptions on the reading process and shed light on the role played by memory processes in reading. However, they were not in a position to examine the effects of interruptions in the context of a normal reading situation, mainly because they used a sentence-by-sentence presentation procedure. In these previous studies,

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the texts were presented one sentence at a time, and when resuming reading the text, the participants were always forced to resume reading at a given point in the text. The main findings were that when the readers were presented with the post-interruption sentences immediately after an interruption, the time spent reading the first post-interruption sentence was inflated compared with an uninterrupted reading situation, whereas overall comprehension was not affected. Generally speaking, our results are well in line with these studies in showing that interruptions have deleterious but limited effects in the context of a reading task, slowing down the reading process without hampering the overall comprehension of the text. Going beyond this, the present study shed light on the way interruptions affect the reading process in a normal reading situation. Our results show that in a normal reading situation, the locus of the interruption costs, in terms of reading time, was exclusively on the pre-interruption text and not on the post-interruption text as reported in previous studies. Our results showed indeed that the processing of the post-interruption text was even *facilitated* by the interruptions, as indicated by a decrease in reading time. Although it came as a surprise, we argue that this result fits very nicely with the Long-Term Working Memory theoretical framework developed by Ericsson and Kintsch (1995). The main conjecture of this framework is that in skilled activities like reading, the end products of processing can be rapidly stored in long-term memory and later rapidly retrieved from long-term memory when retrieval cues are available in short-term working memory. More specifically, we interpret the facilitative effect we observed on the post-interruption text as follows. When the readers return to a text after a long and demanding interruption, a memory trace of the pre-interruption text is stored in long-term memory, but it is not directly accessible as the interruption has wiped out the content of the short-term working memory. If the readers reread the pre-interruption text, the identified words provide retrieval cues that give access to the memory trace stored in long-term memory, but rereading of the pre-interruption text, even for a short period of time, also *reinforce* the memory trace now accessible in long-term memory (see Hyönä & Nurminen, 2006, for evidence of rereading behaviours enhancing memory traces). Then, when processing the post-interruption text, the reinforced memory trace (related to the pre-interruption text) accessible in long-term memory facilitates the integration of the upcoming text, leading to a decrease in reading time compared with the no interruption baseline where the existing memory trace was not reinforced because of repeated readings. Finally, it is worth noting that this result mirrors recent findings obtained by Oulasvirta and Saariluoma (2006, experiment 4). The authors found that when severe time pressure was imposed to the reading task, and rereading of the pre-interruption text was not allowed after the interruption, integration of upcoming text was affected as indicated by a memory loss for the post-interruption text portion. Our result is in line with the findings presented by Oulasvirta and Saariluoma in that they both highlight that, in accordance with the Long-Term Working Memory theory, integration of upcoming information is highly dependent on memory trace related to previously read text accessible in long-term memory.

The second and third goals of the present study were to investigate the effects of background speech and music on the reading process and to examine whether these factors can modulate the effects of interruptions. We obtained the following major results. First, background speech, but not music, led to an increase in overall reading time. This increase in overall reading time was mostly due to readers rereading previously seen words. Second, the comprehension scores in the background speech condition were lower than in the silence condition, but the difference did not reach significance (potentially because of a ceiling effect). Third, we did not find any evidence of modulation of the deleterious effects of interruption in the presence of background speech or music.

The reported findings are in line with the well-established view that background speech greatly disrupts the reading process. The previous studies (Banbury & Berry, 1998; Martin et al., 1988; Oswald et al., 2000; Sorqvist et al., 2010) provided evidence for this disruption by means of off-line measures (i.e. comprehension tests). The present research shows that background speech can also affect on-line processing. Detailed analyses of the readers' eye movement behaviour provided further insight into this disruption: the background speech slowed down the reading process, and this disruption was mostly due to readers rereading previous words within the current sentence and in previous sentences. According to the existing eye movement literature, this is clearly indicative of high-level (post-lexical) integration difficulties (Hyönä et al., 2003; Reichle et al., 2009). Future research could investigate whether lower levels such as lexical processing are also impaired in the presence of background speech. This could be done by examining the fixation durations yielded by low-frequency versus high-frequency words in the presence/absence of background speech.

We stated previously that the disruption of the reading process observed in the presence of background speech is generally given the following explanation: an interference between the semantic processes engaged in the reading task and automatic processing of semantic information in speech (see Marsh et al., 2008, 2009). This explanation is based on the fact that meaningful speech is more disruptive than meaningless speech (Martin et al., 1988; Oswald et al., 2000). Yet, one should note that background speech could also disrupt the reading process at a phonological, as opposed to semantic, level. The interference could also occur at the syntactic level (see Caplan & Waters, 1999). The current study is not in a position to distinguish between these different levels of interference.

Previous studies reported a clear decrease in comprehension performance in the presence of background speech, whereas we failed to find any evidence for such an effect. As stated previously, this discrepancy may be due to a ceiling effect in our measurement of comprehension performance (the comprehension scores were rather high in all conditions). Nevertheless, it should be noted that a comparison of the respective experimental paradigms provides another potential explanation. In the present study, the participants were free to read the texts at their own pace. In contrast, in the previous studies, the time spent reading the texts was always controlled, so that

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participants had exactly the same time to read the texts in the background speech and silence conditions. Yet, our results revealed that readers need substantially more time to read texts when they are exposed to background speech compared with reading in silence. This pattern of results suggests that the experimental paradigms employed in previous studies may have imposed heavy time pressure on the readers, and this time pressure may have led to an overestimation (from an applied point of view) of the deleterious effects of background speech on comprehension performance as they would occur in real life. Note that we do *not* claim that background speech does not have any deleterious effects on comprehension performance. Rather, we interpret our results as suggesting that the deleterious effects of background speech on comprehension performance may be less pronounced than previously thought when reading is self-paced, as in a normal reading situation. It is worth noting that the fact that background speech did not amplify the deleterious effects of interruption is also consistent with this view. The lack of an interaction (interruption costs were even slightly *smaller* in the background speech condition compared with the silence condition) suggests that background speech does not markedly hamper the information stored in long-term memory, as this information seems to be as accurately and rapidly retrieved as in the silence condition when resuming reading the text after the interruption.

Music, compared with background speech, had very little effect on the reading process. The overall reading rate was not affected in particular. Our results are in line with previous studies that showed a lack of effect of instrumental music on the reading process (Furnham et al., 1999; Martin et al., 1988). They also resonate with previous research showing that instrumental music is much less disruptive than vocal music when memorizing a series of words (Salamé & Baddeley, 1989). Nevertheless, a detailed analysis of the eye reader's eye movements yielded a subtle effect. One particular type of rereading behaviour was more frequent in the presence of music than when reading in silence: although the readers did not spend more time rereading previous words within the current sentence, they spent more time rereading words within previous sentences. This suggests that even when the music does not hamper the reading process, it can modify the readers' eye movement behaviour slightly. As pointed out in the introduction, music effects appear very complex, and they are thought to be modulated by a variety of factors (see Jancke & Sandmann, 2010, for a recent review). The primary purpose of our music manipulation was to provide an additional control condition for our background speech condition.

The present study aimed at complementing previous approaches by making use of eye-tracking techniques to gain more ecological validity and further insight into the research issues addressed. Our findings shed light on the way interruption effects might emerge in natural settings. Our results show that in a normal reading situation, the locus of the interruption costs is situated on the text portion already read before the interruption, while processing of the novel text is somehow facilitated. Such a pattern of results is in harmony with the Long-Term Working Memory theory (Ericsson & Kintsch, 1995). Our study also revealed that when individuals read texts at their own pace, as in natural settings, they cope with the interference brought by background speech by slowing down substantially their reading rate and rereading previously read words. Two practical implications can be derived from this last result with regard to work settings where individuals are exposed to background conversations, as in open-office workplaces. First, heavy time pressure should not be imposed to individuals if they are to accurately complete important reading tasks. Second, if the reading task is completed via a computerized interface, individuals should not only have control over their reading pace, but they should also have constant access to previously read information to allow rereading.

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APPENDIX

Example paragraph and comprehension question

For more than eight centuries, poets and writers have been telling stories about King Arthur. From lost legends and scraps of history, from facts and folklore, they have fashioned one of the greatest epics in all literature, full of the world's splendours, heroic loves and spiritual quests. This magical and mysterious world is founded on the figure of an obscure Welsh prince, about whom we know nothing for certain. Arthur may have been last Roman general of Britain, the first of those Welsh guerrilla fighters who defied the English until well into the middle ages. Or a northern prince from Scotland who was later adopted by the Welsh living in Wales. If there was a real Arthur, he lived in the sixth or seventh centuries AD; he may not have even been of royal blood, but he was acclaimed as a hero or leader. That is all that we can say with any confidence about the historical grain of sand in the poetic oyster.

What might King Arthur have been?

- An English prince

- A Roman general

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